

NATIONAL ENGINEERING EXAMINATION - COMPUTER SCIENCE & CLOUD

Course Code: CS-402 | Time: 3 Hours | Max Marks: 100

SECTION A: MULTIPLE CHOICE QUESTIONS (Choose the correct option)

1. What is the primary role of an API Gateway in a microservices architecture?

- (A) Monolithic database query caching
- (B) Reverse proxy, rate limiting, routing, and centralized authentication
- (C) Compiling backend Python source code to bytecode
- (D) Hardware virtualization for CPU registers

2. Which HTTP status code indicates that the client has exceeded its rate limit quota?

- (A) 200 OK
- (B) 401 Unauthorized
- (C) 404 Not Found
- (D) 429 Too Many Requests

3. In asynchronous task processing, which component acts as the message broker between FastAPI and Celery?

- (A) Redis
- (B) PostgreSQL
- (C) Dockerfile
- (D) React Virtual DOM

4. Python decorators are functions that take another function as an argument and extend its behavior.

- (A) True
- (B) False

5. What is the time complexity of searching for an element in a balanced Binary Search Tree (BST)?

- (A) $O(1)$
- (B) $O(n^2)$
- (C) $O(\log n)$
- (D) $O(n!)$

SECTION B: PRACTICAL TERMINAL & DEVOPS ACTIVITY

6. Consider the following Docker execution command:

```
$ docker run -d --name test_redis -p 6379:6379 redis:alpine
```

Explain what the `-d` and `-p` flags signify in this container deployment.

ANSWER KEY

- 1. B
- 2. D
- 3. A
- 4. A
- 5. C